

Annotated Bibliography for SymbolicLongMemorySequences.jl

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July 5, 2026

1 Purpose

This document records the literature foundation for SymbolicLongMemorySequences.jl, a Julia package for synthesizing long-range dependent symbolic sequences. The accompanying BibTeX database is `references/s5_references.bib`. Downloaded PDFs are stored outside this public repository in the private sibling repository described in `DOWNLOAD.mn`; this document and `references/README.md` retain human-readable provenance and interpretation.

The BibTeX file has also been checked with PaperFetch.jl in online check-only mode. Clear DOI/title/author conflicts were corrected in the database, while low-confidence provider failures for books, proceedings, and known local alternatives remain recorded as audit items rather than overwritten.

2 Long-Range Dependence and Self-Similarity

The general mathematical background comes from Samorodnitsky’s survey [1], Pipiras and Taqqu’s monograph [2], and Beran’s statistical treatment [3]. These references distinguish covariance-based long memory, spectral poles, heavy tails, self-similarity, and finite-sample estimation. SymbolicLongMemorySequences.jl uses those distinctions to avoid presenting a finite simulation as proof of asymptotic LRD.

The network-traffic literature, especially Leland et al. [4] and Willinger, Paxson, and Taqqu [5], motivates the practical need for synthetic data that preserves large-scale burstiness. Taqqu, Teverovsky, and Willinger [6], Witt and Malamud [7], and Malamud and Turcotte [8] are used as cautions about estimator dependence, finite scale ranges, trends, and diagnostic choices.

3 Numerical LRD Synthesis

Paxson’s fast spectral synthesis [9] is the direct ancestor of PB1 and the default numerical source for PB2 and PB3. Dieker [10], Coeurjolly [11], Wood and Chan [12], Dietrich and Newsam [13], and Craigmire [14] document exact and approximate Gaussian alternatives. These references support the current SymbolicLongMemorySequences.jl design choice: keep latent numerical generation explicit, so future exact covariance backends can be added without changing the symbolic transformation API.

Sorbye, Myrvoll-Nilsen, and Rue [15] show another scalable path through finite mixtures of short-memory components. Roughan, Veitch, and Abry [16] provide the wavelet and multiscale background behind PB3, which SymbolicLongMemorySequences.jl modifies into a latent-driver plus Markov-regime symbolic generator.

4 Point Processes and Heavy-Tailed Models

Heavy-tailed and fractal point-process generators are represented by Garrett and Willinger [17], Lowen and Teich [18], Thurner et al. [19], and Ryu and Lowen [20]. `SymbolicLongMemorySequences.jl` adapts this line in MB2 through heavy-tailed regime sojourns, and in MB3 through competing symbol-specific renewal streams. Roughan, Yates, and Veitch [21] motivate the validation policy for these methods: finite simulations can miss the intended scale range, so plots must mark interpretation limits.

5 Symbolic and Categorical Applications

Voss [22] is central because it analyzes DNA through one binary indicator sequence per base, avoiding arbitrary scalar encodings of symbols. `SymbolicLongMemorySequences.jl`'s centered one-hot validation diagnostics follow that methodological lesson. DNA references including Peng et al. [23], Li and Kaneko [24], Li, Marr, and Kaneko [25], Buldyrev et al. [26], and Buldyrev [27] motivate symbolic biological use cases and copy/mutation generator families.

Text references include Schenkel, Zhang, and Zhang [28], Ebeling and Poschel [29], Montemurro and Pury [30], Alvarez-Lacalle et al. [31], Altmann et al. [32], Altmann, Cristadoro, and Degli Esposti [33], Tanaka-Ishii and Bunde [34], Debowski [35], and Wieczynski and Debowski [36]. These papers suggest mechanisms such as hierarchy, rare-word clustering, topic persistence, and semantic similarity. `SymbolicLongMemorySequences.jl` does not attempt to model text directly; it provides reproducible synthetic sequences for studying estimators and sequence models under controlled symbolic LRD.

6 Model Details Used by `SymbolicLongMemorySequences.jl`

PB1 combines Paxson-style fGn [9] with rank quantization. PB2 combines fGn streams with an argmax transform inspired by latent Gaussian categorical models [37]. PB3 uses a latent spectral or Haar driver to select Markov regimes, preserving direct local transition control. PB4 uses an intermittent-map source motivated by Provata and Beck [38].

MB1a and MB1b are based on LAMP [39]; `SymbolicLongMemorySequences.jl` adds explicit finite-history and dyadic-bucket variants. MB1c follows additive Markov-chain memory-function work [40, 41, 42]. MB2 and MB3 adapt heavy-tailed traffic and point-process models [17, 20, 18]. MB4 is motivated by Hawkes-process word-occurrence modeling [43]. MB5 is motivated by symbolic DNA growth models [44, 45].

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